Combined SDDP and simulator model for hydropower scheduling with sales of capacity

Martin N. Hjelmeland
Magnus Korpås
Department of Electric Power Engineering
Norwegian University of Science and Technology (NTNU)
Trondheim, Norway
{martin.hjelmeland@ntnu.no}

Arild Helseth
SINTEF Energy Research
Trondheim, Norway

***Abstract*—In following work, we investigate the importance of detailed hydropower scheduling modelling when including sales of capacity, which adds complexity that is not easily incorporated in a Linear Programming (LP) problem. In the proposed approach, we use the profit-to-go function obtained from a Stochastic Dual Dynamic Programming (SDDP) scheduling-model in a Simulator Model, based on Mixed Integer Programming (MIP), and perform detailed simulations. The Simulator Model allows a more complex problem description, than by the LP formulation in the SDDP model. The Simulator Model may therefore be used to give an estimate of the LP approximation, which is used for providing the opportunity cost in short-term hydropower scheduling models or conceivably for making investment decisions.**

For the given case study, the expected profit from selling capacity was 29.2% higher than the linear SDDP Model to the Simulator Model. The overall profit loss was reduced by 0.93%, quantifying the overestimation of profit in the SDDP Model. This illustrates the importance of detailed modelling when considering sales of capacity.

***Index Terms*—Stochastic processes, Power system analysis computing, Optimal scheduling, Hydroelectric power generation**

I. Introduction

With the introduction of a liberalized power market, power producers have been more eager for robust and efficient optimization models for their power scheduling. In Norway the liberalization started in 1991 after the Energy act, stating that power producers were no longer obliged to supply electricity but instead could maximize their profits. In the following year a paper on the EMPS model (EFI's Multi-area Power-market Simulator), which had been developed and used in the Nordic regions since the 1970's, was published [1]. The model mimics the market clearing process and is frequently used by market players to provide price forecasts. A single producer's hydropower strategy need to consider relevant uncertainties, such as inflows and energy prices. The decision process consist of the choice to either store water today and produce tomorrow or vice-versa, i.e. a multi-stage stochastic problem.

Funded by The Research Council of Norway, Project No. 228731/E20

In 1996, the European Union issued an internal electricity market directive (1996/92/EC) to gradually open the electricity markets for all member states. Alongside the political agenda to assemble a common European energy market there has been an even greater push towards a sustainable energy development. With the large-scale entry of non-dispatchable generation, strengthened and possible new market mechanisms are required to attain secure operation of the electricity grid. Due to the flexibility of hydropower, volumes in the balancing markets in Norway have been low, and therefore low incentives for hydropower producers to assess models with multiple market clearing. With an increasing share of renewables and stronger grid connection to other countries, this perception may alter.

Due to the complexity and large computation time detailed model imposes, a clear differentiation has been done for hydropower scheduling. Typically the scheduling comprises a long-, medium- and short-term model depending on the modelling objective [2]. The long-term model may represent an aggregated system model and is e.g. used for expansion planning, price forecasting, and system studies, preferably with large system boundaries [3]. The medium-term model is linked with the long-term model by price and individual water values are calculated for a more detailed part of the larger system. Detailed water values for a river-system may then be used as the opportunity cost as the end-of-horizon valuation of water when running the short-term scheduling, a detailed model often defined as a Mixed Integer Programming (MIP) problem [4–6].

Around 1990, two papers on the Stochastic Dual Dynamic Programming (SDDP) algorithm for multi-stage stochastic optimizations problems was presented in [7, 8]. In the years to come numerous alternations and improvements on the algorithm has been presented. Such as grid modelling in [9], convergence properties in [10] and bidding strategy for a large hydropower portfolio in [11]. Latter years developments include the relations of non-convexities. [12] was treated with use of McCormick envelopes to model the non-convex relationship between power, discharge and water head. Later, [13] used Lagrangian relaxation to assess two procedures on convexification of the cost-to-go function and the non-convex

Authorized licensed use limited to: Norges Teknisk-Naturvitenskap
978-1-5090-1298-5/16/$31.00 ©2016 IEEE
58:30 UTC from IEEE Xplore. Restrictions apply.

components. In [14] a SDDP algorithm with local cuts was proposed in order to capture the non-convexities resulting from a detailed system description and supply of spinning reserves.

The primary frequency reserve (PFR) market in Norway for providing weekly Frequency Containment Reserves - Normal (FCR-N) has especially shown an income potential for hydropower producers during the summer months. With high reservoir fillings, low prices and high inflow, hydropower producers are set to either produce at maximum to avoid spillage or not produce due to allegedly low power prices. The market for PFR in Norway displays many of the same characteristics to other markets and minor changes are need to represent these differences. Comprehensive market details can be found in [15].

In this paper a combined Stochastic Dynamic Programming (SDP)/SDDP method presented in [16] and extended to incorporate capacity reservation in [17], is applied together with a simulator. First, a strategy (represented by the profit-to-go function) is found from a updated version of the SDP/SDDP model described [17], hereby referred to as the Strategy Model. This model allows both delivery of energy to the day-ahead market and sales of capacity to the PFR market. Subsequently the obtained strategy is applied in the MIP-based Simulator Model over the two year period of analysis to find detailed schedules for hydropower stations considering operation in both markets. The Simulator Model has the capability to represent more accurately a non-convex generation-discharge relation, minimum power requirement and head dependencies, if desirable. These details are of particular importance when considering operation in a capacity reserve market, which requires the committed capacity to be spinning. The Simulator Model can therefore act as benchmark to quantify the approximation errors that the Strategy Model impose.

The purpose of the presented approach is to quantify the influence a capacity reserve market has on the medium-term hydropower scheduling, considering accurate modelling of the physical system.

In Section II the Strategy Model and the Simulator Model will be presented. Section III describes a case study on a Norwegian hydropower system followed by results and discussions. Conclusions from the research and case study are finally given in Section IV.

II. Methodology

An overview of the two models can be seen in Figure 1. Inflow series and power prices used in this model has weekly stochastic time stages. In order to scale the model further down, a weekly profile with resolution similar to the PFR market is added, represented by 21 time steps within the week.

It is assumed that the reader has previous knowledge about SDDP, such that details regarding the method are neglected. For interested readers the following articles are suggested [7, 8, 16, 17].

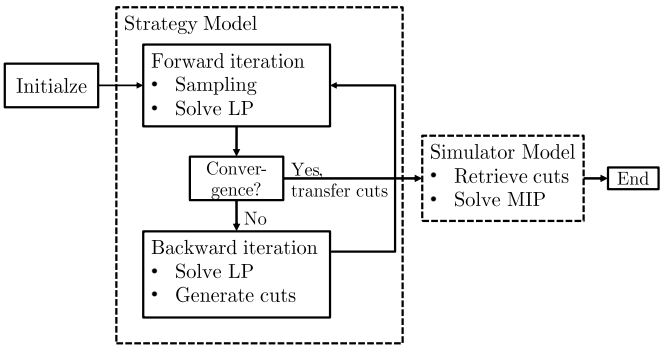


Figure 1. Simplified overview over the models. The Simulator Model is run after convergence is reached in the Strategy Model. Cuts refer to the hyperplanes that the profit-to-go function consists of.

A. Strategy Model

For each time stage, t, and realizations of stochastic variables, ω_t, the hydropower scheduling problem is solved, called the one-stage dispatch problem. The forward and backward iterations of the Strategy Model are carried out iteratively to generate an improved representation of the profit-to-go function until convergence. The LP problem representing the one-stage dispatch problem in the Strategy Model will be discussed in this section. Section II-B will cover the one-stage dispatch problem for the Simulator Model.

$$\alpha_t(\mathbf{x}_{t-1}, \omega_t) = \max\left\{L_t(\mathbf{x}_t, \mathbf{u}_t, \mathbf{c}_t) + \alpha_{t+1}(\mathbf{x}_t, \omega_{t+1})\right\} \quad (1)$$

subject to

$$\mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t = \mathbf{C}\mathbf{x}_{t-1} + \mathbf{D}\mathbf{a}_t(\omega_t) \quad (2)$$

$$\mathbf{E}\mathbf{u}_t = \mathbf{d}_t \quad (3)$$

$$\mathbf{F}\mathbf{u}_t + \mathbf{c}_t \leq \mathbf{b}_t \quad (4)$$

$$\mathbf{u}^{\mathrm{lb}} \leq \mathbf{u}_t \leq \mathbf{u}^{\mathrm{ub}} \quad (5)$$

$$\mathbf{x}^{\mathrm{lb}} \leq \mathbf{x}_t \leq \mathbf{x}^{\mathrm{ub}} \quad (6)$$

$$\mathbf{c}^{\mathrm{lb}} \leq \mathbf{c}_t \leq \mathbf{c}^{\mathrm{ub}} \quad (7)$$

Problem (1)-(7) exhibit the main structure in the one-stage dispatch problem with state variables, $\mathbf{x}_t$, decision variables, $\mathbf{u}_t$, and realization of the stochastic parameter ω_t. $\mathbf{x}_t$ represents, in this model, the amount of water in a reservoir and the state of the power station, while $\mathbf{u}_t$ includes the water discharge, spillage and bypass. The function L_t in the objective function is made up by three major terms; income from selling energy and capacity, and a penalty function. The penalties includes spillage, tank water and start-up costs. Inflow to the reservoir is modelled with a first order autoregressive model, elaborated in [18], such that for extreme cases when the reservoir is empty you might also receive negative inflow. The model can therefore use the very costly tank water to fill up the reservoir. This ensures that relatively complete recourse of the stochastic model is obtained.

One of the characteristic with hydropower scheduling models is the relation between state variables from one time stage to another. This is expressed in constraint (2), where

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 04,2026 at 20:58:30 UTC from IEEE Xplore. Restrictions apply.

the reservoir balance is included. Matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ represent coupling between reservoirs and $\mathbf{E}$ consists of the efficiencies regarding production and discharge in the energy balance constraint (3). Linearized start-up costs, included in (2), were implemented in a similar manner as in [19,20], with a linear approximation of the power stations state (on/off).

The delivery of capacity is incorporated in (4), where $\mathbf{c}_t$ is the amount of capacity available for sale, hence also included in the objective function. The constraints includes a symmetrical delivery of capacity, such that for delivering a certain amount of capacity the power station must be able to regulate both up and down an equal amount. The matrix $\mathbf{F}$ represents the relations between power and discharge and $\mathbf{b}_t$ the power limits.

The delivery of PFR is limited by the turbine governor, such that both the droop setting and size of the power station limits the amount for providing PFR, included in (7). The other variables are bounded by (5) and (6).

Impacts of head variations impose non-convex properties that hydropower scheduling models should consider. Due to its complexity and minor significant impact for the proposed case study, this was currently not treated.

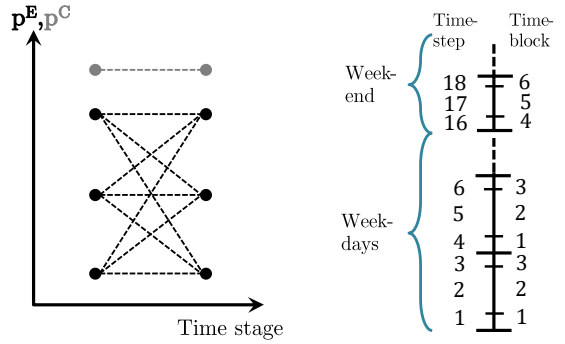


Figure 2. (Left) Illustration of the dynamic recursion between time stages for energy and capacity prices. For the energy price, different price nodes are used with probabilities from going to one price node to another in next time stage. The figure illustrates the underlying modelling of the prices and not the magnitudes. (Right) Illustration of the six different time-blocks for the weekly PFR market, i.e. the capacity for time-block 1 must be available for certain hours for all weekdays, i.e. time step 1,4,7,10 and 13.

1) Price models: In order to ensure a convex function in SDDP, the power price will have to be described in other terms than in the continuous state space. This is achieved by discretizing the power price and using a Markov chain to describe the transition for the power price between time stages. In short, the SDDP model is solved for each of the discretized power prices and transitions between them are used to calculate the expected profit-to-go function, illustrated by the transition between different price nodes from one time stage to another in Figure 2. The PFR price is also described by a deterministic price series in this work, but could be described stochastically as in [21], this would however significantly increase the problem size.

As seen in Figure 2 there are six different periods for providing PFR. If you commit a certain amount in period 1 you must provide the same amount every weekday.

TABLE I. PROBLEM CHARACTERISTICS FOR CASE STUDY.

	Strategy	Simulator
Variables	1 036	2 233 (585 binary)
Constraints	567	2 121
Dispatch problems, per iteration	293 600	5 200

For both markets it is assumed that the hydropower producer is a risk-neutral price-taker. With many actors and the large energy volumes traded on the day-ahead market this is a good assumption. However, the volumes for the PFR market are low, such that an improved representation could be beneficial for this market.

B. Simulator Model

The core distinctions of the Simulator Model is the MIP formulation of the one-stage dispatch problem. This enables a more complex modelling and subsequently more realistic results. The hydropower system for both models are the same, however, the Simulator Model is able to include binary state variables, non-convex power-discharge function and distinct operation from different reservoirs, such that a power station could not generate while discharging from multiple reservoirs.

During the research it was found that one of the main issues with a linear model when considering provision of PFR is that in time periods with low power prices and high PFR prices, the model tended to operate the power station at a minimum level. E.g. the power station was operated at 10 MW in order to provide 10 MW of PFR, even though the technical lower limit of the station was 50 MW. This weakness was significantly improved by reducing the solution space, such that when selling maximum amount of PFR, the power station had to operate above minimum generation limit. This issue was omitted by introducing binary variables in the Simulator Model. Together with the restriction to only generate from one reservoir at the time, this significantly improved the validity of the results given by the Simulator Model.

As Figure 1 illustrates, after the Strategy Model has converged and a solution is obtained, the profit-to-go function, represented by cuts, is utilized in the Simulator Model. Since the profit-to-go function was made for a different optimization problem, a distinct convergence gap is expected. This may be seen as the main drawback of the model, but it ensures a quick and efficient evaluation of the results obtained by the Strategy Model.

III. CASE STUDY

The case study is based on a hydropower system under construction in south-west of Norway, shown in Figure 5. Power station 1 and 2 has installed capacity of 13.8 MW and 400 MW, respectively. The price series for capacity reserves are given by the actual prices from 2013 and 2014. A simulation with the EMPS model from 2013 gave the price series for energy, such that the capacity and energy prices were coupled in time.

In order to get a honest comparison between the two models, all Simulator Model runs utilized the scenarios for inflow and

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 04,2026 at 20:58:30 UTC from IEEE Xplore. Restrictions apply.

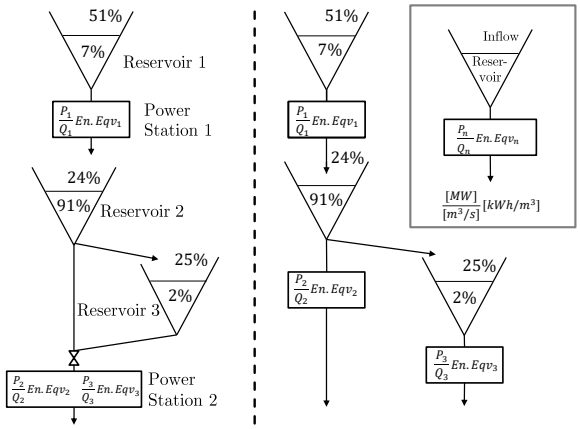


Figure 5. System used in the Simulator Model (left). Simplified system used in the Strategy Model (right). Size of reservoir and inflow given as percentage of the whole system. Power station 2 has two operating points in regard to which reservoir it is operating from. P_n, Q_n and $En.eqv_n$ refer to the power-discharge relations for the given power station and reservoir n.

spot price that were sampled in the last forward iteration by the Strategy Model.

The simulation was carried out with 50 scenarios in the forward iteration of the Strategy Model and hence the same amount in the Simulator Model. The sampling of 50 scenarios could favourably have been increased, but due to long computation time and promising in-sample stability this was not done. The power prices were discretized into 7 price nodes while the number of inflow samples in the backward iteration were 8. There were carried out 15 iterations of the Strategy Model and a final simulation with the Simulator Model.

The number of variables and constraints in the one-stage dispatch problem for the two models is displayed in Table I, together with the total number of problems. It can be seen that the Simulator Model has a significant increase in both variables and constraints for the problem described in the case study.

A. Results and Discussion

All simulations were carried out on a Dell Latitude E7240 with an Intel Core i7-4600U processor with a 2.7 GHz clock rate and 16 GB RAM. The CPU time for the Strategy Model was 11 h and 4 min and 41 min for the Simulator Model.

TABLE II. EXPECTED VALUES FOR THE SIMULATOR MODEL, GIVEN IN PERCENTAGE WITH THE STRATEGY MODEL AS BENCHMARK.

Generation	-1.0	Profit	-1.48
Energy Profit	-0.99	End Value of Water	3.26
Capacity Profit	-29.18	Total Profit	-0.93

Some selected results, in percentage with the Strategy Model as benchmark, are given in Table II. It is shown that the expected total profit, referring to the sum of profits and the end value of the water, was reduced by 0.93%. The profit from sales of capacity was reduced by 29.2%. This significant reduction from sales of capacity comes as a result of the tightened system description with minimum generation and operating reservoir constraints, and it shows how vital the system description is when considering provision of PFR.

Figure 3 shows the convergence of both models. It can be seen that there is some uncertainty concerning the gap in the Strategy Model, as a new sampling of inflow and power price is carried out for each iteration. As expected an convergence gap is observed in the Simulator Model, in percentage calculated to 0.68%.

The duration curve depicted in Figure 4 shows how generation, for the entire hydropower system, is shifted from less high power output to increased output at around 150 MW. This is first seen because of the binary constraints, but also as a result of the detailed power-discharge modelling, where the power station 2 has beneficial properties around this point. Also, due to the minimum generation limit in the Simulator Model, no generation beneath this limit is obtained, as opposed to the Strategy Model where this is observed with some power outputs of around 50 MW.

The proposed method does not improve the profit-to-go strategy used by short-term models, as in [12–14], but instead gives a rapid benchmark of how far away the linear approximation is from the true value. This can be very efficient for making investment decisions, where additional electricity

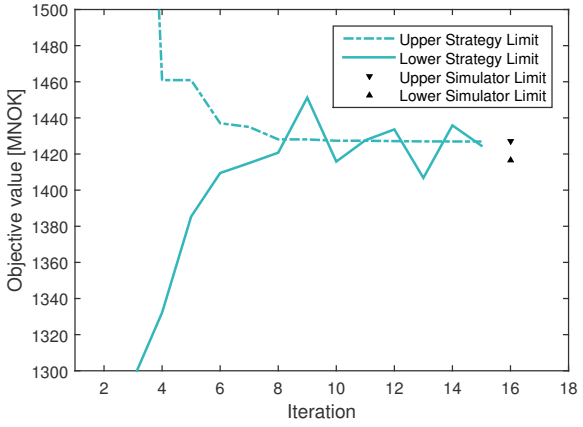


Figure 3. Convergence of both the Strategy and Simulator Model.

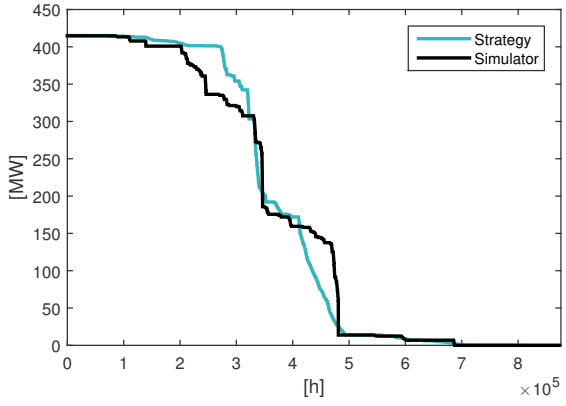


Figure 4. Duration curve from the Strategy and Simulator Model.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 04,2026 at 20:58:30 UTC from IEEE Xplore. Restrictions apply.

markets are included. An example could be to either invest in a Pelton turbine that provides good generation efficiency over a large amount of different discharge values, such that delivery of reserves are fairly cheap, or to invest in a Francis turbine with a higher maximum efficiency but with a higher cost of providing capacity reserves.

IV. Conclusion

A realistic approach for medium-term hydropower scheduling operating in both day-ahead and capacity reserve markets has been formulated and tested. The Simulator Model proposed has a detailed description of the power-discharge function and minimum generation for hydropower. When modelling the accurate power-discharge function, generation will give more realistic results regarding the unit's operating point. The minimum generation restriction also provides realistic results concerning how the hydropower units are operated. Following by the reduced flexibility in the system a profit reduction was observed, adjusted with the end value of water, by 0.93%.

By generating the profit-to-go function from the Strategy Model and applying it in the same system with above-mentioned modifications, a prompt solution is obtained, dependent on the system size and simulation period. It can be observed that the Strategy Model overestimated the value of selling capacity by around 29%. Even though this value is case-specific, it demonstrates the importance of detailed modelling when considering sales of capacity.

Further work should consist of generating an improved representation of the profit-to-go function that is to be used by the Simulator Model and hence reducing its convergence gap. Nevertheless, the finding from the case study with 0.68% gap demonstrate encouraging properties in terms of finding the optimal solution. Improvements of the representation of the capacity market should also be empathized in regard to validity of prices and conceivable volumes.

Nomenclature

ω_t Stochastic parameter for time stage t.

$\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}$ System specific matrices. Representing coupling between reservoirs, relations between water discharge, energy and capacity.

$\mathbf{a}_t(\omega_t)$ Inflow as a function of a realization of the stochastic parameter ω_t.

$\mathbf{c}_t$ Capacity for sale.

$\mathbf{u}_t$ Decision variable. Representing discharge of water.

$\mathbf{x}^{\mathrm{lb}}, \mathbf{u}^{\mathrm{lb}}, \mathbf{c}^{\mathrm{lb}}$ Lower bounds for the variables.

$\mathbf{x}^{\mathrm{ub}}, \mathbf{u}^{\mathrm{ub}}, \mathbf{c}^{\mathrm{ub}}$ Upper bounds for the variables.

$\mathbf{x}_t$ State variable. Representing reservoir volume and state of a power station.

References

[1] O. Botnen, A. Johannesen, A. Haugstad, S. Kroken, and O. Frystein, "Modelling of hydropower scheduling in a national/international context," *Hydropower '92*, 1992.

[2] O. B. Fosso, A. Gjelsvik, A. Haugstad, B. Mo, and I. Wangensteen, "Generation scheduling in a deregulated system. the norwegian case," *IEEE Transactions on Power Systems*, vol. 14, no. 1, pp. 75–81, 1999.

[3] O. Wolfgang, A. Haugstad, B. Mo, A. Gjelsvik, I. Wangensteen, and G. Doorman, "Hydro reservoir handling in Norway before and after deregulation," *Energy*, vol. 34, no. 10, pp. 1642–1651, 2009.

[4] E. K. Aasgård, G. S. Andersen, S. E. Fleten, and D. Haugstvedt, "Evaluating a stochastic-programming-based bidding model for a multireservoir system," *IEEE Transactions on Power Systems*, vol. 29, no. 4, pp. 1748–1757, 2014.

[5] O. B. Fosso and M. M. Belsnes, "Short-term hydro scheduling in a liberalized power system," in *Power System Technology, 2004. PowerCon 2004.*, vol. 2, Conference Proceedings, pp. 1321–1326 Vol.2.

[6] S.-E. Fleten and T. K. Kristoffersen, "Short-term hydropower production planning by stochastic programming," *Computers & Operations Research*, vol. 35, no. 8, pp. 2656–2671, 2008.

[7] M. V. F. Pereira and L. M. V. G. Pinto, "Multi-stage stochastic optimization applied to energy planning," *Mathematical Programming*, vol. 52, no. 1-3, pp. 359–375, 1991.

[8] M. V. F. Pereira, "Optimal stochastic operations scheduling of large hydroelectric systems," *International Journal of Electrical Power & Energy Systems*, vol. 11, no. 3, pp. 161–169, 1989.

[9] B. G. Gorenstin, N. M. Campodonico, J. P. da Costa, and M. V. F. Pereira, "Stochastic optimization of a hydro-thermal system including network constraints," *IEEE Transactions on Power Systems*, vol. 7, no. 2, pp. 791–797, 1992.

[10] A. B. Philpott and Z. Guan, "On the convergence of stochastic dual dynamic programming and related methods," *Operations Research Letters*, vol. 36, no. 4, pp. 450–455, 2008.

[11] B. C. Flach, L. A. Barroso, and M. V. F. Pereira, "Long-term optimal allocation of hydro generation for a price-maker company in a competitive market: latest developments and a stochastic dual dynamic programming approach," *IET Generation, Transmission & Distribution*, vol. 4, no. 2, pp. 299–314, 2010.

[12] S. Cerisola, J. M. Latorre, and A. Ramos, "Stochastic dual dynamic programming applied to nonconvex hydrothermal models," *European journal of operational research : EJOR*, vol. 218, no. 3, (1.5), pp. 687–697, 2012.

[13] F. Thome, M. V. F. Pereira, and M. Granville, Sergio Fampa, "Non-convexities representation on hydrothermal operation planning using sddp," working paper, 2013, accessed: 07.10.2015. [Online]. Available: http://www.psr-inc.com.br/portal/psr/publicacoes/

[14] H. Abgottspon, K. Njalsson, M. A. Bucher, and G. Andersson, "Risk-averse medium-term hydro optimization considering provision of spinning reserves," *International Conference on Probabilistic Methods Applied to Power Systems 2014 (PMAPS 2014), Durham, UK, 2014.*

[15] ENTSO-E, "Network code on load-frequency control and reserves," 2013. [Online]. Available: https://www.entsoe.eu/

[16] A. Gjelsvik, M. M. Belsnes, and A. Haugstad, "An algorithm for stochastic medium-term hydrothermal scheduling under spot price uncertainty," *Power System Computation Conference (PSCC)*, p. 7, June 28 - July 2nd, 1999 1999.

[17] A. Helseth, B. Mo, M. Fodstad, and M. N. Hjelmeland, "Co-optimizing sales of energy and capacity in a hydropower scheduling model," in *PowerTech, 2015 IEEE Eindhoven*, June 2015, pp. 1–6.

[18] A. Gjelsvik, B. Mo, and A. Haugstad, *Long- and Medium-term Operations Planning and Stochastic Modelling in Hydro-dominated Power Systems Based on Stochastic Dual Dynamic Programming*, ser. Energy Systems. Springer Berlin Heidelberg, 2010, book section 2, pp. 33–55.

[19] G. Warland, A. Haugstad, and E. S. Huse, "Including thermal unit start-up costs in a long-term hydro-thermal scheduling model," *Power System Computation Conference (PSCC)*, 2008.

[20] A. Helseth, A. Gjelsvik, B. Mo, and U. Linnet, "A model for optimal scheduling of hydro thermal systems including pumped-storage and wind power," *IET Generation, Transmission & Distribution*, vol. 7, no. 12, pp. 1426–1434, December 2013.

[21] A. Helseth, M. Fodstad, and B. Mo, "Optimal medium-term hydropower scheduling considering energy and reserve capacity markets," *IEEE Transactions on Sustainable Energy*, pp. 1–9, 2016. [Online]. Available: http://dx.doi.org/10.1109/TSTE.2015.2509447

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on November 04,2026 at 20:58:30 UTC from IEEE Xplore. Restrictions apply.